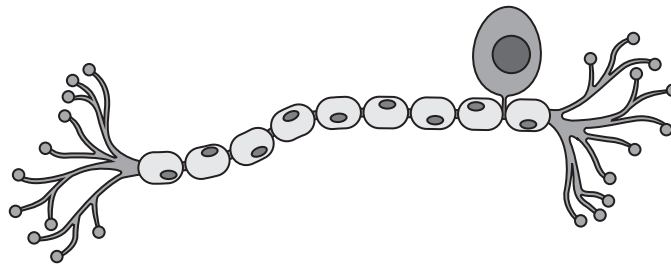


1 Neurones transmit impulses between receptors and effectors.

The diagram shows a neurone.

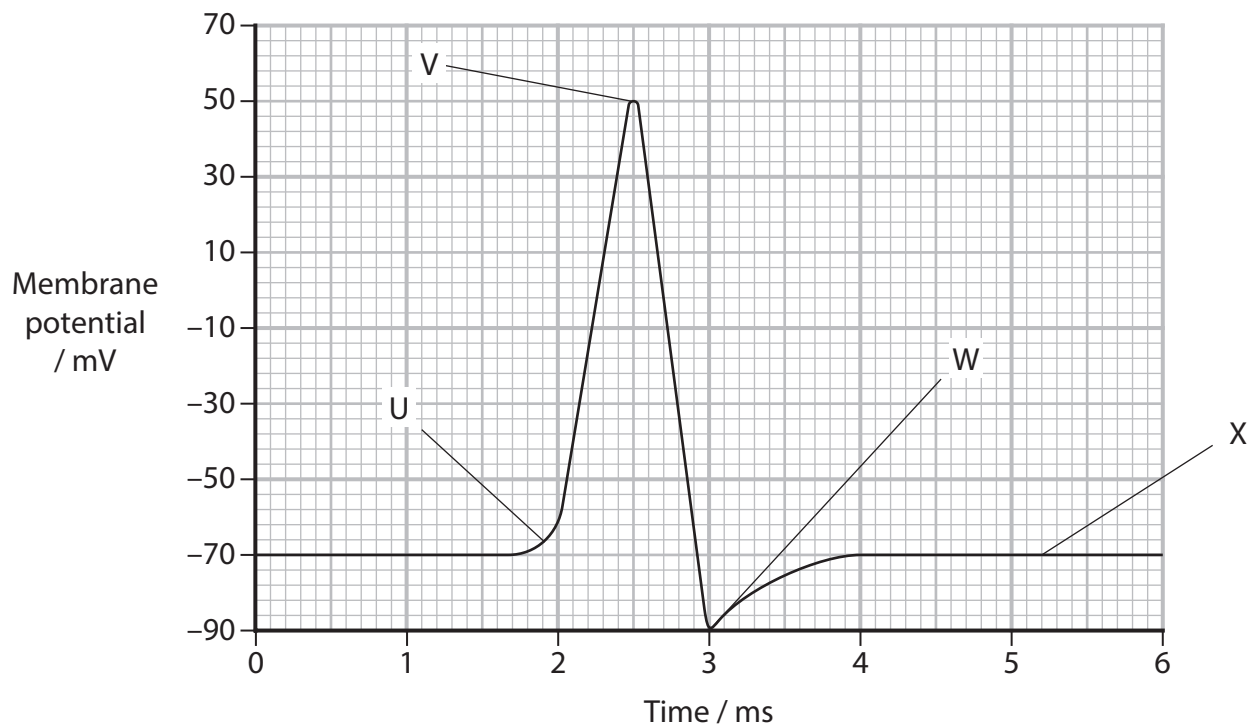


(a) Draw an arrow on the diagram to show the direction of a nerve impulse.

(1)

(b) Nerve impulses travel along axons due to changes in membrane potentials.

The graph shows an action potential in a neurone.



(i) Which is the resting potential for this neurone?

(1)

- ☐ **A** -90 mV
- ☐ **B** -70 mV
- ☐ **C** +48 mV
- ☐ **D** +70 mV

(ii) Which letter on the graph shows where the neurone is hyperpolarised?

(1)

- ☐ **A** U
- ☐ **B** V
- ☐ **C** W
- ☐ **D** X

(iii) Which is the difference in membrane potential between points V and W on the graph?

(1)

- ☐ **A** 40 mV
- ☐ **B** 50 mV
- ☐ **C** 90 mV
- ☐ **D** 140 mV

(c) Give **two** differences between the **functions** of a motor neurone and a sensory neurone.

(2)

(Total for Question 1 = 6 marks)